

Package ‘LongSubTest’

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Type Package

Title LongSubTest

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Description Change-plane testing methods including the supremum and the weighted average of score test are proposed for longitudinal data.

License GPL (>=2)

Imports Matrix,mnormt

Repository github

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gam.init	<i>Function to select initial gamma values spanning its space</i>
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Usage

```
gam.init(n.initials, Z, lb.quantile, ub.quantile, ss=1)
```

Arguments

n.initials	Number of gamma values.
Z	The grouping variables.
lb.quantile	The lower quantile specified for $Z\% \gamma$ initial values.
ub.quantile	The upper quantile specified for $Z\% \gamma$ initial values.
ss	A positive integer with n.initials/ss indicating how many sets of gamma are chosen, default to 1.

Value

A matrix. A set of gamma values selected for defining subgroup.

Examples

```
n = 200
m = 4
p1 = 2
p2 = 1
p3 = 3
alpha = rep(1, p1)
beta = rep(1, p2)/2
gamma = c(1, seq(-1,1,length.out = p3-1))
rho = 0
set.seed(100)
data = generate_data(n, m, alpha, beta, gamma, rho)
K = 1000
qlb = 0.1
cols = apply(data$U, 2, var) != 0
Gamma = gam.init(K, data$U[,cols], lb.quantile=qlb, ub.quantile=1-qlb, ss=1)
```

generate_data

Function for generating longitudinal data from a change plane model.

Usage

```
generate_data(n, m, alpha, beta, gamma, rho)
```

Arguments

n	A constant. The sample size.
m	A constant. The number of time point.
alpha	A vector. The true parameter for baseline covariates.
beta	A vector. The true parameter denoting the heterogeneous effect of the subgroup.
gamma	A vector. The true parameter for grouping variables.
rho	The strength of correlation among covariates.

Value

A list.

- Y - A $n * m$ matrix. The response.
- X - A $(n * m) * p$ matrix. The baseline covariates.
- Z - A $(n * m) * q$ matrix. The baseline covariates.
- U - A $(n * m) * r$ matrix. The grouping variables.

Examples

```
n = 200
m = 4
p1 = 2
p2 = 1
p3 = 3
alpha = rep(1, p1)
beta = rep(1, p2)/2
gamma = c(1, seq(-1,1,length.out = p3-1))
rho = 0
set.seed(100)
data = generate_data(n, m, alpha, beta, gamma, rho)
```

LongSubTestST

The score test for the change-plane model with longitudinal data.

Usage

```
LongSubTestST(data, Gamma, B = 1000, K = 1000, qlb = 0.1, seed = 1)
```

Arguments

data	A list, including Y ($n * m$ response), X ($(n * m) * p$ predictors), Z ($(n * m) * q$ predictors), U ($(n * m) * r$ grouping variables).
Gamma	A matrix for initial gamma values. If not given then it will be automatically generated based on the data.
B	A constant. Number of bootstrap samples. Default is 1000.
K	A constant. The number of the initial gamma values. Default is 1000.
qlb	A constant. The lower quantile specified for $Z * \gamma$ initial values. Default is 0.1.
seed	A constant. The number of seeds for generating the initial gamma values. Default is 1.

Value

A list.

- `tn0` - The value of test statistic.
- `pval` - The p-value of the test.
- `time` - Running time.

References

Subgroup testing for longitudinal data.

Examples

```
n = 200
m = 4
p1 = 2
p2 = 1
p3 = 3
alpha = rep(1, p1)
beta = rep(1, p2)/2
gamma = c(1, seq(-1,1,length.out = p3-1))
rho = 0
set.seed(100)
data = generate_data(n, m, alpha, beta, gamma, rho)
fit <- LongSubTestST(data)
```

LongSubTestWAST	<i>The weighted average of the score test for the change-plane model with longitudinal data.</i>
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Usage

```
LongSubTestWAST(data, family = 1, B = 1000, seed = 1)
```

Arguments

data	A list, including Y ($n \times m$ response), X ($(n \times m) \times p$ predictors), Z ($(n \times m) \times q$ predictors), U ($(n \times m) \times r$ grouping variables).
family	The corresponding family. family = 1 denotes the Gaussian family, and family = 2 denotes the binomial family. Default is 1.
B	A constant. Number of bootstrap samples. Default is 1000.
seed	A constant. The number of seeds for generating the initial gamma values. Default is 1.

Value

A list.

- tn0 - The value of test statistic.
- pval - The p-value of the test.
- time - Running time.

References

Novel subgroup testing for longitudinal data.

Examples

```
n = 200
m = 4
p1 = 2
p2 = 1
p3 = 3
alpha = rep(1, p1)
beta = rep(1, p2)/2
gamma = c(1, seq(-1,1,length.out = p3-1))
rho = 0
set.seed(100)
data = generate_data(n, m, alpha, beta, gamma, rho)
fit <- LongSubTestWAST(data)
```

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